Is there a mass above which quantum superposition stops? Registered readings of mass and distance, the 2026 interference record and a prediction dated to 2035

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Abstract

Standard quantum mechanics has no mass at which a superposition stops: large objects look classical because of decoherence, the environment that keeps recording where they are. Collapse models (Ghirardi–Rimini–Weber, CSL, Diósi–Penrose) add such a limit by hand. The framework says that a threshold exists: below a certain mass level the principal GCD of the point of observation and the GCD of the diffraction that generates the object become equal, and the object spreads over several positions (see the box above). The box gives no number. This deposit tries to give the threshold a number, or to say clearly why it cannot. A second part tests two later remarks of the author: the limit cannot be fixed as a mass, because a calculation of the «fractional distance» is missing; and large distant bodies stay fixed because their size makes up for the distance. It was announced in deposit 2.64 (DOI 10.5281/zenodo.23610071).

Is there a mass above which quantum superposition stops? What the experiments say and a prediction that expires in 2035

Student version of MM record 2.66 (DOI 10.5281/zenodo.23643759), written with an AI assistant from the archived text; the archived text remains on Zenodo.

English title of the record: Is there a mass above which quantum superposition stops? Registered readings of mass and distance, the 2026 interference record and a prediction dated to 2035.

Ugo Lanciano, October 2026. Licence CC BY 4.0 for the texts, MIT for the code.

Deposit 2.64 (mass as a contained rhythm) had left one question aside. It concerned the mass threshold of the quantum world, "treated in a separate work with a prediction dated to 2035". This is that work. It can be read on its own. It has two parts, done with the same method:

- the threshold as a mass (22 October 2026);
- the role of distance (25 October 2026, §§3.3, 4.6, 5.5). This part was added before upload, after two remarks of the author.

In brief

The question

Small things behave in a strange way. Take an atom, or a molecule. Send it through a set of slits, that is, of thin openings. It does not go through one slit or the other. It goes through all of them at once. On the screen it leaves the stripes of a wave, like the waves of water that pass through two gaps in a dam. Physicists call this **superposition**: the object is spread over several positions at the same time.

Large things, instead, are never seen like this. A chair is always in one place. Where does the change happen? And why?

What today's physics says

Standard quantum mechanics has no mass at which the change happens. It accounts for the classical look of large objects with **decoherence**. The environment constantly "looks" at a large object: air molecules, light, heat. So the stripes are washed out before anyone can see them. Think of a drawing made with sand on a table that everyone keeps bumping into. The sand stays, but the drawing is erased at once.

Some physicists think there is more. They think there is a true limit, present even in a perfect vacuum, beyond which a superposition cannot survive. These are the **collapse models**. The best known ties the limit to gravity (Diósi and Penrose).

What the author's framework says

The author's framework, MM, says that such a limit exists. At the top of every record there is a summary, "MM in brief". There it says that, when an object falls below a certain mass level, two quantities of the framework become equal. They are the "principal GCD" of the point of observation and the "GCD of the diffraction" that generates the object. Then the object spreads over several positions, and we observe it as probability. GCD stands for greatest common divisor. We will explain it in section 2.

That summary gives no number for the threshold. This work tries to give it one. Or it tries to say clearly why it cannot.

After the first part, the author added two remarks. First: the limit cannot be fixed as a mass, because the calculation of a «distanza frazionale» ("fractional distance") is missing. Second: at an adequate distance even a large object could take on quantum features. Large distant bodies, instead, stay fixed, because their size makes up for the distance. The second part tests these remarks.

What we did

In each of the two parts we did three things.

1. **The data.** We downloaded 20 papers from arXiv, the public archive of physics papers. We listed what the experiments have seen. There are nine experiments with quantum behaviour. They go from single atoms to a vibrating crystal of 16 micrograms. In addition there are the limits on collapse models. For the second part we used 22 more sources:
2. decoherence measured while changing the separation and the number of scattered photons;
3. quantum correlations over distances from 20 cm to 1,203 km;
4. the experiments on "Wigner's friend";
5. the diameters of stars measured with interferometry;
6. the threshold of the naked eye.
7. **Readings.** We translated the thesis into precise criteria: ten for the mass, eight for the distance. Each says which physical quantities stand for the terms of the framework. And so it says where the threshold lies. We **registered** all of them, with the outcome rules, before doing any calculation. Registering means writing in a file and fixing it with a fingerprint. So it cannot be changed afterwards. Each translation is a choice of the AI; they are not the author's words.
8. **Tests.** For the readings that remain there is a prediction with a deadline of 31 December 2035. Anyone can apply its rules. For the distance there is also a test with data that already exist. It was registered before reading the sources that decide it.

The five main results

- **For distance, the limit is not a mass alone.** There are two readings of the fractional distance that hold. In both, the limit is a ratio. It is the size to be told apart against the "cell" of the copies that do the telling, together with the number of copies. (The readings of the first part with a mass threshold at the Planck scale remain possible. But they are not tested.) The two readings have a similar form. That they are a single rule of the framework is a proposal.
- For the **state** of an object the copies are the photons and molecules that the environment carries away. The cell is the scale at which one copy tells two positions apart. For light it is about half a wavelength. Below that scale each copy takes away only a small part of the coherence. This is why a crystal of 16 micrograms stays quantum over $2 \cdot 10^{-18}$ m. To fix it, about 10^{34} thermal copies would be needed.
- Take the way in which a **distant** observer reads an object. Here the cell is the diffraction cell of the observer's

aperture, at the distance of the object. Two stars of almost the same mass fall on opposite sides. The disc of Sirius B (1.02 solar masses, 2.64 parsec) is 20-60 times smaller than the finest cell of today's interferometers. This is a calculation from its radius. τ Ceti (0.78 solar masses, farther, 3.65 pc) and 61 Cyg A (0.67-0.74, 3.49 pc), instead, are resolved. This means that their disc is measured. With this instrumental classification the sample does not support a mass limit of the reading, of mass alone, that grows with distance.

- The reading "what matters for the state of an object is how far the observer is" **does not follow from the data**. Fullerene molecules are heated, with the same apparatus and the same observer. The interference goes from 47 % to 7 %, then to zero. The cause is photons that nobody collects.
- **Three readings do not follow from the data.**
- The threshold as an internal acceleration compared with the galactic scale a_0 (a reading from the author's book).
- The threshold at the mass of the proton or of the electron.
- a_0 written as a mass.

In 2026 clusters of about 7,500 sodium atoms (172,000 atomic mass units) showed interference. They are held together by accelerations 26 orders of magnitude above a_0. That means a hundred million billion billion times larger. A caesium atom has 132 times the mass of the proton. It kept a superposition of 1.9 micrometres for 70 seconds.

- **Five readings remain possible, but they cannot be tested today:**
- two thresholds near the Planck mass: 22 micrograms, and 15.4 micrograms in the "two walls" form;
- a threshold at the mass of the observer: 70 kg;
- two thresholds of the Diósi and Penrose type.

Take an object heavier than about $3 \cdot 10^{-22}$ kg. No experiment has yet put its centre of mass into a superposition as large as the object itself. In today's interference experiments the effect of gravity is at most $4.2 \cdot 10^{-10}$ of what would be needed to see it.

- **The reading with the clock of deposit 2.40 falls exactly on the 2026 record** (ratio 1.24). But it is not a general threshold. It coincides with the design condition of grating interferometers, the Talbot condition. Atom interferometers, where the superposition is created by laser pulses, go beyond it by seven orders of magnitude, that is, by ten million times. It describes a condition of the apparatus, that is, of the reading, not a property of the object.
- **The dated prediction is registered**: time 22 October 2026, 13:00:30 UTC; deadline 31 December 2035. A program applies its rules and passes 45 tests out of 45. The heaviest planned experiment is the space mission MAQRO. In the most optimistic study scenario it would reach 0.095-0.35 of the effect of gravity. Within the limits of the published study, at most 0.05. The most likely outcome in 2035 is "not assessable".

No experiment has met a threshold, so far. All the data are explained by standard quantum mechanics with decoherence. The reading "no intrinsic threshold, the threshold is decoherence" is compatible with them. The two rules of the fractional distance that hold coincide with known physics. Known physics here is decoherence, and the wave optics of the light that carries the image. The framework admits them. An experiment can put them to the test. But it cannot tell them apart from standard physics.

How to read this text

- Section 1 summarises what physics already knows, and what the author's framework had already said. On a first reading it can be skimmed.
- Section 2 explains the thesis without formulas.
- Section 3 says how the thesis was tested.
- Section 4 gives the results.
- Section 5 gives the dated prediction.
- Section 6 says what is not claimed.

The second part, on distance, is in section 1 (last paragraphs), in §2.2, in §3.3, in §4.6 and in §5.5. Technical words are explained the first time they appear. And they are explained again in the glossary. Numbers such as 2.38, 2.40, 2.48 and 2.64 refer to earlier works of the author. They are deposited on Zenodo. The list with the DOIs is at the end.

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1. What is already known

Quantum mechanics has no mass threshold

Nothing in the theory stops a superposition when the object grows. Large objects look classical because of decoherence (Joos and Zeh 1985; Zurek 2003; Schlosshauer 2019). Every collision with a gas molecule or with a photon carries away a trace of where the object is. And the stripes disappear.

For large objects this happens in very short times. Take a speck of dust spread over a hundredth of a millimetre. Decoherence takes about 10^{-31} s in air. In the best laboratory vacuum it takes about 10^{-14} s (Schlosshauer 2019, Table 1). For a decoherence threshold the environment decides, not the mass.

Collapse models put in a threshold "by hand"

Ghirardi, Rimini and Weber (1986) proposed a model, called GRW. There is also a continuous version, called CSL (Pearle 1989; Ghirardi, Pearle and Rimini 1990). These models add to quantum mechanics a small random process. The process localises objects, that is, it fixes them in one place. For one atom it is rare. For many atoms together it is fast. Two numbers fix it: a rate λ and a length r_C.

- The values proposed by Ghirardi, Rimini and Weber ($\lambda = 10^{-16}$ s^{-1} at r_C = 10^{-7} m) are not excluded yet.
- Larger values are excluded. They are excluded by the emission of X-rays in germanium detectors and by mechanical experiments (Carlesso and others 2022; the review by Carlesso and Donadi 2025).

Diósi (1987, 1989) and Penrose (1996) tie the collapse to gravity. A superposition of two positions of a mass lasts about ħ/E_G. Here E_G is the gravitational energy of the difference between the two arrangements of the mass. The symbol ħ is the reduced Planck constant, the basic number of quantum mechanics.

- Take the version in which there is no number to be adjusted by hand. This version of the model is excluded by an underground experiment at Gran Sasso (Donadi and others 2021).
- There is a version with a radius R_0 over which the mass of each nucleus is spread. The same kind of measurement requires $R_0 > 2.54 \cdot 10^{-10}$ m (Majorana Demonstrator 2022, as corrected by its erratum).

Older still is the work of Károlyházy (1966).

How "quantum" an experiment is

Nimmrichter and Hornberger (2013) proposed a number: the **macroscopicity** μ. The larger μ is, the more the experiment restricts any modification of quantum mechanics of the collapse-model type.

What the experiments have seen

Here is the inventory of the data, with the values read in the papers. The letter u is the atomic mass unit: $1.66 \cdot 10^{-27}$ kg, about the mass of a proton. Δx is the separation between the two positions. T is the time.

	experiment	object	mass	separation Δx	time T
E1	Pedalino and others, Nature 649, 866 (2026)	clusters of 5,000-10,000 sodium atoms, about 8 nm across, in a three-grating interferometer	143,000-197,000 u (mean 172,000 u, $2.9 \cdot 10^{-22}$ kg)	grating period 133 nm	6.1 ms between two gratings
E2	Fein and others (2019), second hand	molecules of about 2,000 atoms	over 25,000 u	not read	not read
E3	Bild and others, Science 380, 274 (2023)	a vibration mode of a sapphire crystal, prepared in a "cat state"	effective mass 16.2 μg	$2.1 \cdot 10^{-18}$ m	under 84 μs
E4	Delić and others (2020)	silica nanoparticle cooled to the ground state	$2.8 \cdot 10^{-18}$ kg	zero-point motion only	7.6 μs
E5	Rossi and others (2025)	silica nanoparticle	$1.2 \cdot 10^{-18}$ kg	coherence 73 pm	—
E6	Tomassi and others (2026)	silica nanoparticle, expanded thermal state	$2.0 \cdot 10^{-18}$ kg	coherence 1-7	260 μs

				pm	
E7	Whittle and others (2021)	a vibration mode of the LIGO mirrors	effective mass 10 kg	zero-point motion only	—
E8	Panda and others, Nature Physics 20, 1234 (2024)	one caesium atom held in an optical lattice	133 u	1.9 μm	up to 70 s
E9	Kovachy and others (2015), second hand	rubidium atoms	87 u	54 cm	not read

A few words for the table. "Second hand" means read in another paper, not in the original. A "cat state" is a superposition of two clearly different states of a large system. It is like Schrödinger's famous cat. The "zero-point motion" is the minimum trembling that remains even at a very low temperature. LIGO is the large detector of gravitational waves. The prefixes: nm is a billionth of a metre, pm a thousandth of a nm. For time, μs is a millionth of a second, ms a thousandth.

E1 holds the record of macroscopicity: μ = 15.45. It is about one order of magnitude, that is, ten times, above the previous record. Its stripes have visibility 0.10 and 0.08. They follow the quantum prediction, not the classical one.

Here is a fact of the data, with no reading. Above about $3 \cdot 10^{-22}$ kg no experiment has put an object into a superposition as large as the object itself. Heavier objects behave in a quantum way only over very small distances, from 10^{-11} to 10^{-18} m.

What is planned

- The space mission **MAQRO** (Gasbarri and others 2021; launch foreseen in 2034) studies particles of 10^7 to 10^{11} u in free fall. The free fall lasts up to 100 s. Its feasibility study gives visible stripes up to 40 s and particles up to 400 nm.
- The authors of E1 announce a next step at a million atomic mass units. And then, in a vertical interferometer, another factor of a hundred in mass and in coherence time.
- Carlesso and others (2022) estimate what is needed to reach the rate of Ghirardi, Rimini and Weber. According to the estimate, about 10^7 u, a superposition of 180 nm and 20 s are needed.

Distance, in the experiments (second part)

The separation against the scale of the copies.

- In an atom interferometer a single scattered photon removes all the contrast when the two paths are about half a wavelength apart. It is "the resolving power of an ideal Heisenberg microscope". Beyond that, the contrast comes back at intervals (Pfau and others 1994, Chapman and others 1995, as reported by Kokorowski and others 2001).
- With many photons the contrast falls as $\exp(-\kappa^2 d^2/2)$. At fixed d it falls almost exponentially with the mean number of photons (Kokorowski and others 2001, $\bar{n}$ from 0.9 to 8.2).
- Fullerenes (carbon molecules shaped like a ball) are heated before an interferometer. The stripes are lost: 47 %, 29 %, 7 %, 0 % at 0, 3, 6 and 10.5 W. The cause is thermal photons that go off into space. The loss follows the wavelength of the photons compared with the separation (Hackermüller and others 2004).
- With a residual gas the visibility falls as $\exp(-p/p_0)$. It agrees with the theory of decoherence, with no numbers adjusted by hand (Hornberger and others 2003).
- **The copies carried away count even if nobody collects them.**

The distance between the parts, and from the source of the settings. There are correlations beyond the Bell bound. The Bell bound is the maximum that two objects can have in common if each one follows only local rules. Examples:

- between atoms at 398 m ($S = 2.221 \pm 0.033$);
- between electron spins in two diamonds at 1.3 km (2.42 ± 0.20);
- between photons sent from a satellite to stations 1,203 km apart (2.37 ± 0.09). Here the settings are chosen by the light of stars hundreds of light years away and of quasars billions of years old;
- entanglement, that is, a quantum link, between two mechanical oscillators 20 cm apart;
- a "quantum eraser" at 55 m (visibility 0.95) and at 144 km (0.75-0.78, with a link attenuated by 33 dB).

They are different set-ups, not a controlled series. They show quantum behaviour at every separation tried. But none of

them changes only the distance between a system and whoever records it.

Wigner's friend. It is a thought experiment, today actually carried out. One observer, the "friend", measures inside a closed laboratory. An outside observer treats the whole laboratory as a quantum system. In the experiments the records of the friends are coded in paths of photons, or in a few qubits of a quantum computer. The outside observers see correlations that would not be allowed if the outcomes of the friends were absolute (with locality and without superdeterminism). The experiments are: Proietti and others 2019, $S = 2.416 \pm 0.075$; Bong and others 2020; Zeng and others 2025; a preprint of 2026. No experiment has a friend with a macroscopic mass.

The reading of distant objects.

- The light of a distant star is coherent over an aperture smaller than about λ/θ (van Cittert-Zernike theorem). Here θ is the angular diameter. This is why interferometers measure the diameters of stars from the fringes. Examples: α Cen A 8.502 ± 0.038 mas, τ Ceti 2.022 ± 0.010 mas, 61 Cyg A 1.775 ± 0.013 mas. The mas is a thousandth of an arcsecond: a very small angle.
- The naked eye, adapted to the dark, sees a flash of 54-148 photons at the cornea (Hecht, Shlaer and Pirenne 1942). It sees a single photon just above chance (0.516 ± 0.010, Tinsley and others 2016).

These data concern the **reading**, not the **state** of the objects.

What the author's framework had already said

- The author's book NUMERI (v119, §§[01404]-[01422]) places the threshold in a ratio. It is the ratio between the strength of the bond that holds an object together and its mass. It is a quantity with the dimensions of an acceleration. The book notes that its form is close to the galactic scale $a_0 = 1.2 \cdot 10^{-10}$ m/s². And it adds that this is a convergence of form, not a proven identity. Here it is reading T0.
- The status report of 26 August 2026 (version 2.9, file NON_SALDATO.md) had found two things. The threshold, used as a criterion, predicts nothing beyond standard decoherence. And a version of it as a rate of collapse is excluded by the caesium atom of E8. It proposed to read it as a threshold on the reading, not on systems. Here it is reading T4.
- The "two walls" invariant of the status reports 2.9-2.18. The Compton wavelength of a mass and its Schwarzschild radius cross at $m = m_P/\sqrt{2} = 15.4$ μg. Here it is reading T1b.
- Deposit 2.38 counts the "layers" between the Planck scale and a mass m as $\ln(m_P/m)$: 44.0 for the proton, 51.5 for the electron (the value of the electron is in 2.38). Here they are readings T1 and T1c.
- Deposit 2.40 reads the time of a clock as the geometric mean of its two light sides. Here it is reading T2.
- Deposit 2.48: in the framework, probability passes through the Born rule. The exponent 2 of that rule is taken from quantum mechanics, not derived.
- For the second part:
- deposit 2.36.2 says: "distance is the measure of the fragmentation of a register". And the fragmentation behaves like a Brownian motion that deposits 1/r (the source of the potential);
- deposit 2.62 says that a reading is a projection "in fractions" on the ruler of whoever reads. Two readers share the common fractions. The object they share appears in a few tens of copies. With an object of several values the common part grows in steps (working notes of 25 October 2026);
- deposit 2.50 tested a translation of «venuta in essere e distanza di diffrazione» ("coming into being and diffraction distance") for dark energy. None of 18 combinations came within 0.02 of Ω_Λ.
- The expression «distanza frazionale» ("fractional distance") has no rule in the earlier material.

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2. The thesis, explained

2.1 The framework, in a few lines

In the author's framework the physical world is made of relations. Space and time have only one logical element in common. The author calls it their **greatest common divisor (GCD)**. It is like the numbers 3 and 5, which have only 1 in common. The local world arises when a relation is counted through this GCD. Seen by an observer inside, the multiples appear as time and the fractions as space. The observed objects are linked to the observer by a "diffraction", in fractions, of the counting of whoever observes.

2.2 What the thesis says about quantum mechanics

The summary "MM in brief" published on every record says this. When an observed object is large enough, its mass

fixes it in the counting of the point of observation. Then we see it in one place and at one instant. When it falls below a certain mass level, two GCDs become equal, as in a mirror. One is the principal GCD, that of the point of observation. The other is the GCD of the diffraction that generates the object. Then the object no longer occupies a single point. It spreads over several possible positions. And we observe it as probability. The summary adds that the idea of a passage from quantum to classical that depends on mass is close to that of Penrose and Diósi.

Three things in this text are **readings by the AI**, approved by the author for publication:

- "principal GCD" = the GCD of the point of observation;
- "GCD of the diffraction" = that of the diffraction that generates the object;
- the threshold = the point where the two become equal.

The exact words of the author are in a separate file of the package.

Two later remarks of the author (25 October 2026), told in the words of physics.

- **The first.** The mass limit cannot be determined, because a calculation of the «distanza frazionale» ("fractional distance") is missing. At an adequate distance even a macroscopic object could take on quantum features.
- **The second** answers an objection: the Moon is seen sharp from 384,000 km. Distant bodies such as the Moon are large enough to stay fixed, even with a high fractional distance. A direct test would be to look at a single stone on Mars with the naked eye, without any instrument.

The author calls the first remark a suspicion to be evaluated.

2.3 What has to be decided to give the threshold a number

- **What "quantum behaviour" means, as something measured.** We fixed two observables. Q1: stripes of interference of the centre of mass that differ from the classical prediction. Q2: a state of motion that no classical description allows between distinguishable positions. It is shown by a negative Wigner function.
- **Mass alone is not enough.** Collapse models depend also on the separation Δx between the two positions. And they depend on the time T for which the superposition lasts. A threshold must say m*(Δx, T): a mass that depends on separation and time.
- **It must be intrinsic.** A threshold of the framework should be there even without an environment. Where decoherence already destroys the stripes, a threshold cannot be seen. And this must be said.
- **What the two GCDs are, in physical quantities.** The framework does not say it. Each translation of section 3 makes a choice. The choices are the AI's. **An outcome hits that choice, not the author's premise.**
- **What "distance" is, and for whom.** Two things must be kept apart. The **state** of the object: is the object itself in superposition? And the **reading**: how does it appear to an observer who looks at it from far away? The data on the Moon, on the stars and on the eye concern the reading, never the state. Each reading of §3.3 says which of the two it speaks about.

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3. How the thesis was tested

3.1 The inventory (Level 0)

Twenty papers were downloaded from arXiv. For each experiment we noted mass, separation, time, environment, outcome and the page where they are written. Before any calculation the experiments were divided into three groups:

- **decisive**: E1, E3, E8;
- **supporting**, which cannot exclude a reading on their own: E2 and E9, read second hand; E5;
- **not decisive**, which show no superpositions beyond the zero-point motion: E4, E6, E7.

For E3 the **effective mass** is a choice of the authors of the experiment. It is the mass assigned to a vibration mode. And it depends on how its amplitude is defined. For this reason E3 decides a reading only on one condition. The outcome must stay the same when the effective mass changes by a factor of 100, up or down.

3.2 The ten readings (Level 1)

For every pair (reading, experiment) a program computes a ratio r.

- $r < 1$ means: the experiment is on the quantum side of the threshold.
- $r > 1$ means: it is on the side where the reading says "fixed in one place".

The outcome rules were fixed before the calculation:

- **excluded** (it is the registered label; in this text we say "does not follow from the data"): a decisive experiment shows quantum behaviour with $r > 10$;
- **compatible, put to the test:** no exclusion, and a decisive experiment has $0.1 \le r \le 10$;
- **not testable today:** no exclusion, and all decisive experiments have $r < 0.1$.

The factor 10 covers the conventions of order one:

- the 2π between h and ħ;
- a factor 2 between the conventions of Diósi and of Penrose;
- Δx taken equal to the period of the grating.

Example: if the true threshold were 3 times higher or lower, the rule would not change the outcome.

reading	principal GCD read as	GCD of the diffraction read as	threshold	origin
T0	— (the book's criterion)	—	internal acceleration (bond / mass) below a_0	book NUMERI
T1	the GCD at the origin (Planck scale)	the GCD of the object	$m^* = m_P = 22$ µg (layers $\ln(m_P/m) = 0$)	AI, on 2.38
T1b	— (the two walls: the Compton wavelength of the object equals its Schwarzschild radius)	—	$m^* = m_P/\sqrt{2} = 15.4$ µg	book NUMERI; status reports 2.9-2.18
T1c	the GCD of our own scale (proton; variant: electron)	the GCD of the object	m^* = mass of the proton (or of the electron)	AI, on 2.38
T1d	the mass of the observer	the mass of the object	$m^* = 70$ kg	book NUMERI
T2	the light side of the observation, cT	the diffraction side of the object, h/(mc)	the geometric mean $\sqrt{(hT/m)}$ equal to Δx: $m^* = hT/\Delta x^2$	AI, on 2.40
T3a	the energy resolution of the observation, ħ/T	the gravitational energy E_G of the two positions, mass spread over the whole object	$E_G \cdot T = \hbar$	AI, on Penrose
T3b	as T3a	as T3a, with the nuclei spread over $R_0 = 2.54 \cdot 10^{-10}$ m	$E_G \cdot T = \hbar$ (Diósi-Penrose at the experimental limit)	AI, on Diósi-Penrose
T4	the environment that records the position	—	no intrinsic threshold: decoherence rate × T = 1	status report 2.9
T5	the background, a_0	the acceleration $G\,m/\Delta x^2$ with which the object reads its own mass	$G\,m/\Delta x^2 = a_0$	AI, on the book

A few explanations. The **Planck mass** m_P is about 22 micrograms. It is the mass of a speck of dust. The **Compton wavelength** h/(mc) is a length linked to every mass. The **Schwarzschild radius** is the radius at which a mass would become a black hole. The scale a_0 is the small acceleration of galaxies, about $1.2 \cdot 10^{-10}$ m/s².

For T3a and T3b we need the gravitational energy of a uniform sphere displaced by Δx. It is that of Howl, Penrose and Fuentes (2019, equation 23). For T4 the rate of collisions with the residual gas is that of Romero-Isart (2011, equation 27). Every constant is written in the file registered on 22 October 2026. None is chosen after the calculation.

What was known before writing the readings. The registered predictions were written after estimates made from memory by the AI. They say so openly: they are **contaminated**. Contaminated means: something of the result was already known. All ten outcomes agree with those estimates. One numerical closeness was known before the calculation. It is $m_P/\sqrt{2} = 15.4$ µg against the 16.2 µg of E3. By the rule on the effective mass it has no weight. It was registered as a

coincidence without structure.

3.3 The second part: readings of the fractional distance

The same three steps. First, the inventory of the book, of the earlier deposits and of the experiments (Level 0, section 1). Second, eight readings with their outcome rules. They were registered on 25 October 2026 at 12:59:58 UTC, before any calculation. Third, the tests (Level 2, §5.5).

There are two observables:

- **Q**, the state of the object, as in the first part;
- **L**, the reading of an observer. Does the object appear fixed within its own size? Or does it appear spread over the diffraction figure of the observer's aperture? In that figure each photon arrives at a random point.

The outcome rules are those of §3.2, with one addition. Take a reading that says that a quantity is constant. In a decisive experiment that quantity must not change by more than a factor 2 (or by more than 5σ).

reading	the fractional distance is	how mass enters	observable
D0	the physical distance, with no rule of its own (standard quantum mechanics)	as in the first part	Q
D1	the separation between the two positions, measured in the scale at which a copy carried away by the environment tells them apart (for light, its wavelength λ: u = Δx/λ); after N copies the coherence is s(u)^N, and the object is fixed when the copies make its position common to the records of the environment (the threshold of deposit 2.62, tolerance 10^{-6})	through the number of copies per unit time (size, temperature, pressure)	Q
D2a	the distance between the object and whoever records it, alone; the copies that nobody collects do not count	a mass limit that grows with that distance	Q
D2b	the distance from the nearest record, of any kind; it reduces to D1 (object not isolated) or to D4 (isolated)	—	Q
D4	an informational distance: the fraction f of the copies of an outcome that reaches the observer (f = 0 for a closed laboratory, as for Wigner's friend)	a macroscopic friend produces many copies: f = 0 requires total isolation	Q, for the observer
D5a, D5b	the fragmentation of deposit 2.36.2: its copies carry the image (D5a), or its deposited share w = $Gm/(2Lc^2)$ fixes the reading when w ≥ w* (D5b, with w* = 1, 1/2, 1/(16π))	through the number of copies, or through w	L
D6	the diffraction cell of the copies (the light) in the observer's aperture a, at the distance L of the object, in units of the object's size D: F = 1.22 λL/(aD); fixed if F ≤ 1; for F > 1 the reading is the figure of the aperture, the same for every object smaller than the cell	only through the size (D6m: a mass limit that grows with L)	L
D6b, D6c	with N copies collected the centre of the figure is known to δ/√N (D6b); below a threshold of copies there is no reading at all (D6c; for the eye 54-148 photons at the cornea)	through the number of copies	L
D7	the fractionation grows with the travel time of light from the observer, by 1 + κ L/(c t_0) with κ = 4.05 (a law tested on galaxy clusters in the working notes)	the mass limit shifts by the same factor	Q

To understand D6, think of a telescope in space that looks at a star. There the air does not disturb. If the star is large enough, or close enough, the telescope sees its disc. If it is too small or too far away, you see only a little spot. The spot always has the same shape, whatever star is there. It is the figure of the telescope's aperture, produced by the diffraction of light. F says how small the object is compared with the smallest cell that your instrument can tell apart at that distance.

"Diffraction cell", "parity" and "becoming equal" are bridges chosen by the AI. In D6 the phrase of the summary "the two GCDs become equal" is read like this. The reading takes on the grain of whoever observes (F > 1: the figure is that of the aperture, not of the object). The phrase of the second remark «così grandi da parificare i due MCD» ("large enough to equalise the two GCDs") is read like this. The size of the object is a multiple of that grain (F ≤ 1). D4 and D6 are ideas of the AI. D1 gives a rule to the reading "the threshold is decoherence" (T4).

4. What comes out

4.1 All the outcomes

reading	decisive number	registered outcome	in words
T0	E1: sodium bound at $1.26\cdot10^{16}$ m/s², r = $1.0\cdot10^{26}$; E8: r = $2.2\cdot10^{31}$	excluded	does not follow from the data
T1c	E8: caesium, r = 132; E1: r = $1.7\cdot10^{5}$ (electron variant: $2.4\cdot10^{5}$ and $3.1\cdot10^{8}$)	excluded	does not follow from the data
T5	E3: r = $2.0\cdot10^{27}$, and at least $2\cdot10^{23}$ changing the effective mass by 100 times	excluded	does not follow from the data
T1	decisive experiments at most r = $1.3\cdot10^{-14}$ (E1); E3 does not decide (0.0074-74)	not testable today	alive, not testable
T1b	as T1 (E3: 0.011-105)	not testable today	alive, not testable
T1d	E3: $2.3\cdot10^{-10}$	not testable today	alive, not testable
T3a	r ≤ $4.2\cdot10^{-10}$ (E8)	not testable today	alive, not testable
T3b	r ≤ $9.3\cdot10^{-14}$ (E1)	not testable today	alive, not testable
T2	E1: r = 1.24	compatible, put to the test	holds as a condition of grating interferometers, not as a general threshold (§4.3)
T4	E1: r = 0.29	compatible, put to the test	coincides with standard quantum mechanics

4.2 The readings that do not follow from the data

- **T0, the internal acceleration of the book.** In E1 there are clusters of sodium atoms. In each cluster the atoms are held together by accelerations of about $1.3\cdot10^{16}$ m/s². They are 26 orders of magnitude above a_0. And yet the clusters interfere. The same holds for the caesium atom of E8. If the threshold were the internal acceleration compared with a_0, these objects should be fixed in one place. They are not.
- **T1c, the threshold at our own scale.** A caesium atom (132 proton masses) and clusters of 172,000 u behave in a quantum way. So the threshold is not at the mass of the proton, nor at that of the electron.
- **T5, a_0 as a mass.** For the vibrating crystal of E3 this reading would give a ratio of $2\cdot10^{27}$. Changing the effective mass by a factor of 100 does not save it. The link that the book suggests between galaxies and the quantum world does not carry over in this form.

What falls. Each translation of the two GCDs into a physical quantity falls. These translations are choices of the AI. The author's premise does not fall.

4.3 The reading with the clock of deposit 2.40 (T2)

The reading. In deposit 2.40 the time of a clock is the geometric mean of its two light sides. The geometric mean of two numbers is the square root of their product. For 4 and 9 it is 6. Here one side is the diffraction side of the object, h/(mc). The other is the light side of the observation, cT. Then the space that the observed point occupies in the time T is $\sqrt{(hT/m)}$. The threshold is where this space equals the separation Δx: $m^* = hT/\Delta x^2$.

What the data say. In E1 we take Δx equal to the grating period. And we take T equal to the flight time between two gratings. We find r = 1.24. The record experiment sits exactly on this threshold. With ħ in place of h the ratio would be 7.8. And it would exceed 10 as soon as the true separation were 13 % larger than the period. So the data prefer h. With h the outcome stays the same for separations up to about 2.5 times the period.

Why it is not a general threshold. The condition $m = hT/\Delta x^2$ is exactly the **Talbot condition**. It is the condition with which a three-grating interferometer is designed. The first grating localises the object at the scale of the period. The stripes form only if the wave spreads at least that much. The idea of T2 was written after reading this in the paper of E1.

It is declared as contamination.

In quantum mechanics the coherence of a free object grows as $\hbar T/(m\sigma_0)$. Here σ_0 is how localised the object was at the start. T2 is the case $\sigma_0 \approx \Delta x$, up to numerical factors. Where the superposition is created by laser pulses, quantum mechanics goes far beyond T2. Take the atom interferometer of Kovachy and others, with 54 cm and at most 3 s. There $r \geq 2.1 \cdot 10^7$ (an experiment read second hand). In E8, in the phase in which the separation of 1.9 μm is built (0.267 ms), r = 4.5.

Outcome. With the registered rules T2 remains "compatible, put to the test". But the reading "$m^* = hT/\Delta x^2$ is the threshold of quantum behaviour" **does not follow**. A narrower reading remains: T2 describes the condition of grating interferometers. It is a threshold on the reading, not on the object. T2 is not identical to quantum mechanics. It is a particular case of it, which quantum mechanics exceeds.

4.4 Decoherence (T4)

Take the collisions with the residual gas in E1. If the gas is air at the stated pressure, they remove about a quarter of the coherence (r = 0.29). They remove about two thirds if the gas is hydrogen (r = 1.08). Hydrogen is the usual residual gas of an ultra-high vacuum. It is the reading "threshold on the reading, not on systems", proposed in August. It coincides with standard quantum mechanics by construction. The second part gives it a rule, D1 (§4.6). The rule uses the separation measured in the scale of the copies, and the number of copies.

4.5 The readings that cannot be tested today

- **T1, T1b, T1d: thresholds at the Planck scale or above.** No experiment has put an object heavier than about $3 \cdot 10^{-22}$ kg into a superposition as large as itself. The crystal of E3 (16.2 μg effective) does not decide. The reason is that its effective mass depends on a choice of the authors. The closeness between 15.4 μg (T1b) and 16.2 μg remains a coincidence without structure.
- **T3a, T3b: the readings of the Diósi and Penrose type.** In today's experiments $E_G \cdot T/\hbar$ is at most $4.2 \cdot 10^{-10}$. Let us do a cross-check. For the crystal of E3 the formula of T3b puts the boundary at $R_0 \approx 2\text{-}4 \cdot 10^{-17}$ m. It is the same order of magnitude as a re-analysis of that experiment (Fadel 2023; pointed out by the verifier and not read in full).

An illustration (outside the protocol: it is not a test). Take a silica object that must spread as much as its own size in free flight.

- After 1 ms the T2 condition is met up to $3.3 \cdot 10^{-21}$ kg ($2 \cdot 10^6$ u, 14 nm). The effect of T3a becomes of order one at $6.6 \cdot 10^{-14}$ kg (3.9 μm).
- After 100 s the two values are $3.3 \cdot 10^{-18}$ kg (0.14 μm) and $6.6 \cdot 10^{-17}$ kg (0.39 μm).

A test of a threshold linked to gravity therefore needs long times and very localised initial states. Or a spreading faster than free flight. Or space.

4.6 The second part: distance

All the outcomes agree with the predictions registered together with the readings. But those predictions were contaminated. The AI knew the main papers before writing them.

reading	decisive number	outcome
D0	correlations beyond the Bell bound at every separation tried, but no experiment changes only the distance from whoever records	not testable today (compatible)
D1	one copy erases the contrast at $\Delta x = \lambda/2$ (r = 1.00); the heated fullerenes and the residual gas follow the theory of decoherence	compatible and put to the test (standard decoherence)
D2a	same apparatus, same observer, only the molecules heated: visibility 47 % → 7 % (factor 6.7) → 0	does not follow from the data
D2b	—	no outcome of its own (it reduces to D1 and D4)
D4	friends coded in paths of photons or in a few qubits: violations of the Bell-type and	compatible; not testable today as a

	"local friendliness" inequalities	rule (no data on the leak of copies)
D5a	predicted width of the image about 2 rad, against stars read as discs of $4\cdot10^{-8}$ rad ($r = 4.9\cdot10^{7}$)	does not follow, with the bridge "the copies of 2.36.2 carry the image"
D5b	the Moon, read sharp, has $w = 7.1\cdot10^{-14}$; with $w^* = 1, 1/2, 1/(16\pi)$ it would not be fixed	does not follow for these w^*; with w^* free, not testable today (only $w^* \leq 7.1\cdot10^{-14}$)
D6 (sharp boundary at F = 1)	HD 177724, a star with $F \approx 1.8$ at the longest baseline of CHARA in the K′ band, has a diameter measured to 2 % (0.897 ± 0.017 mas; found after the calculation)	does not follow from the data
D6g (graded form, written after the calculation)	the reading depends only on F, continuously: for a disc the visibility of the fringes is $\lvert 2J_1(x)/x\rvert$ with $x = 3.83/F$; the Moon (F = 0.01 for the naked eye), α Cen A, τ Ceti and 61 Cyg A (F = 0.19-0.92 on the scale of CHARA)	compatible; it is wave optics, added after the calculation, not a blind test
D6m	Sirius B (1.018 $M_\odot$, 2.64 pc) too small by a factor 57 for CHARA in K′; τ Ceti (0.783 $M_\odot$, 3.65 pc) and 61 Cyg A resolved	does not follow, with the instrumental classification (decided in §5.5, block R)
D6b	positions of unresolved stars known to 1/1500 of the cell of Hipparcos; the number of copies is required by the model, not counted	compatible; not testable today
D6c	a stone of 10 cm on Mars seen with the naked eye at the minimum distance: $3\text{-}6\cdot10^{-9}$ photons per second at the cornea, one every 5-10 years, against 54-148 in 0.1 s	compatible: no reliable visual reading
D7	the correction is $3\cdot10^{-26}$ in a laboratory, $1.7\cdot10^{-15}$ at Mars, $1.3\cdot10^{-9}$ at α Cen	not testable today

CHARA is a large interferometer for stars, in California. Hipparcos was a satellite that measured the positions of the stars. $M_\odot$ is the mass of the Sun. The parsec (pc) is a unit of distance for stars, a little more than three light years.

What falls.

- The reading "for the state of an object only the distance from the observer counts". By changing only the environment, with the same apparatus and the same observer, the stripes change.
- The bridge "the copies of the fragmentation of deposit 2.36.2 carry the image". And the three values of w^* tried for its deposited share. With w^* free, D5b remains not testable today. Deposit 2.36.2 with those copies reads the potential 1/r. This deposit is not touched.
- A mass limit of the reading, of mass alone, that grows with distance, in the sense of block R (§5.5). With the registered instrumental classification the sample does not support it.
- The sharp form of the cell. Below the cell the size can still be read, gradually.

What holds, and what it says.

- **For the state of the object (D1).** The fractional distance is the separation measured in the scale at which the copies of the environment tell the two positions apart. The boundary is reached when the copies make the position common to the records of the environment: $N(m, T)\cdot(-2 \ln s(u)) = \ln(1/\delta)$. About 40 copies are needed at a sixth of the scale, 10,000 at a hundredth. Take the crystal of E3, with the thermal photons of a cryostat at 10 mK. There about $2\cdot10^{34}$ are needed.
- **For the reading (D6g).** The fractional distance is the observer's cell at the distance of the object, measured in units of the object's size. The reading depends on it continuously. The many copies of a bright body fix its position inside the cell. This is what the second remark of the author says.

In both, mass enters only through the number of copies and the size. In these two readings there is no single mass limit. Two objects of the same mass can fall on opposite sides: two stars; a hot molecule and a cold one. The data test decoherence and optics. That the boundary is the point where the object becomes the common part of the records is the framework's reading.

The stone on Mars. Seen with the naked eye, a stone on Mars sends too few copies for any reading. One photon arrives every 5-10 years, against a threshold of 54-148 photons in a tenth of a second. A rock would have to be 30-40 km wide to reach the threshold. And it would still be mixed with the light of Mars. Mars sends $4.6\cdot10^{15}$ times more. The test cannot be done with the eye. And not because of a limit of technology: the copies are missing. That for such an observer the stone "casts no shadow in the record" (deposit 2.62) is the framework's reading, not a result of the count. For the state

of the object the distance that counts is not the one from the observer. It is the informational one: that no copy of the position goes out towards anyone.

A proposal, not a result. D1 and D6g have the same form. In both there is a size to be told apart against the cell of the copies of whoever reads, with the number of copies. They could be a single rule of the framework. This would be close to an earlier idea of the author. According to that idea, the fractionation of an object is read from its distance of diffraction from the observer. The data do not say so. The quantities are different.

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5. The dated prediction (Level 2)

What is registered, and when. The prediction and its program were registered on 22 October 2026 at 13:00:30 UTC. That time is the time limit of the targets. Only papers that came out afterwards count. Just before, there was a search in the arXiv data (June-October 2026). It found no experiments in the region that gives information. The prediction concerns T1, T1b, T3a and T3b.

Two corrections were registered a few minutes later. At that time no target could yet exist. The first concerns the parameters of MAQRO (§5.3). The second came after the verification. The rules valid today are those of the second correction and of the program that applies them. If text and program do not agree, the program holds.

5.1 Which papers count

A paper counts if it meets all these conditions:

- its first version on arXiv (or in the journal, if that comes first) has a date after the time limit. The date does not go beyond 31 December 2035;
- it reports an experiment, not a proposal;
- it gives the mass, the separation, the time, and when needed the size and the material;
- it gives the observable (Q1 or Q2) with its significance.

Confidence levels and p-values are converted into standard deviations with fixed formulas. The target is the first paper, in order of date, with a ratio of at least 0.1.

5.2 The rules

- **Planck block (T1, T1b), r = m/m*.** The prediction: above m* no quantum behaviour of the centre of mass. If quantum behaviour is seen at 3σ or more:
- "compatible, put to the test" for $0.1 \le r < 1$;
- "tension" for $1 \le r \le 10$;
- "excluded" for $r > 10$, whatever the visibility.

"Favourable" only if all these things hold:

- quantum behaviour is not seen at 3σ or more;
- $r \ge 1$;
- the stripes vanish above m* (fraction of the expected visibility $f + 2\sigma_f < 0.2$);
- in the same apparatus, with a mass of at most m*/2, the stripes are there ($f - 2\sigma_f > 0.5$), with no known cause given by the authors.

For a vibration mode the outcome must hold for every effective mass within a factor of 100.

- **Gravity block (T3a, T3b), r = E_G·T/ħ.** The prediction: the superposition is reduced by a factor exp(−r), beyond the known decoherence. If quantum behaviour is seen at 3σ or more:
- "compatible" for $0.1 \le r \le 1$;
- "tension" for $1 < r \le 10$;
- "excluded" for $r > 10$. "Excluded" also if the observed fraction of visibility is clearly above what the reading allows ($f - 2\sigma_f > 10 \cdot \exp(-r/2)$).

"Favourable" if:

- $r \geq 1$;
- the observed fraction of visibility is clearly below one ($f + 2\sigma_f < 0.8$);
- the fraction lies in the band allowed by the reading with a factor 2 on r ($\exp(-2r)/3 \leq f \leq 3\cdot\exp(-r/2)$);
- the authors give no known cause for the loss.

In this block the rules on f come before those on σ. The reason is that T3 predicts a reduced interference, not an absent one.

- The definition that holds for all these rules is the program.
- If by 31 December 2035 no paper reaches $r \geq 0.1$, the block is "not assessable".
- A favourable outcome for T3 would not distinguish the author's framework from Penrose or from Diósi-Penrose. The formulas are theirs. It is written so in the prediction.

5.3 What the planned experiments would reach

We use the parameters of the MAQRO study: grating period 50 nm, silica at 1,850 kg/m³.

project	mass	r for T3a (Δx = 50 nm / 100 nm)	r for T1b
MAQRO, 100 s (study scenario)	10^{11} u	0.095 / 0.35	$1.1\cdot10^{-8}$
MAQRO, 100 s (study scenario)	10^{10} u	0.0087 / 0.029	$1.1\cdot10^{-9}$
MAQRO within the limits of the feasibility study (400 nm, 40 s)	up to about $3.7\cdot10^{10}$ u	at most 0.05	—
E1, next step at 1 MDa and 25 m/s	10^{6} u	$2.2\cdot10^{-11}$	$1.1\cdot10^{-13}$
E1, vertical interferometer	$1.7\cdot10^{7}$ u	$3.6\cdot10^{-8}$	$1.9\cdot10^{-12}$
the target of Carlesso and others for the rate of Ghirardi, Rimini and Weber	10^{7} u	$3.3\cdot10^{-7}$	$1.1\cdot10^{-12}$

Take the most optimistic MAQRO scenario, with a separation of two grating periods. Only this scenario enters the region in which the rules give an answer ($r \geq 0.1$). And it enters only as "compatible, put to the test": far from a clear outcome. The experiments with atoms or molecules below 10^7 u stay below 0.1 for both blocks. They stay there even with T up to 1,000 s and Δx up to 1 m.

A correction to know about. Before the correction the AI had estimated from memory a ratio of about 6 for MAQRO. With the right parameters it is 0.095-0.35. So the test written with that estimate ("r between 1 and 10") does not hold. It is reported as it is. The tables printed before the correction use the wrong parameters of MAQRO.

5.4 What the prediction does not cover

- Take a new limit on Diósi-Penrose that does not come from interferometers (X-rays, heating). Such a limit concerns the radius R_0 of T3b. It does not concern T3a, which is a criterion without dynamics.
- T2 and T4. T4 has no predictions different from quantum mechanics. T2 would have some. In MAQRO it would differ from quantum mechanics. But atom interferometers already exceed it. For T2 no dated prediction was made.

How to apply it. Anyone can use the program of the prediction. You give it the numbers of the paper. The program applies the rules and the admission criteria. Its 45 tests include the exact edges r = 0.1, 1 and 10. They also include the counter-examples raised by the verifier.

5.5 The tests of the distance readings (second part)

They were registered on 25 October 2026 at 13:08:06 UTC. That was before downloading the two primary sources that decide block R (13:25:09 UTC). However, the values that decide it were known from the abstracts. And the expected outcome was written. So it is the application of a registered rule to primary sources read afterwards. **It is not a blind test.**

- **Block R, decided: is the limit of the reading a mass or a size?** Suppose a mass limit that grows with distance. It would have to lie above 1.02 solar masses at 2.64 pc. Indeed Sirius B is not fixed in the reading. Its predicted disc, 0.029 mas, is below a tenth of the finest cell of CHARA in K′, 0.163 mas. And it would have to lie below 0.78 at 3.65 pc. Indeed τ Ceti (2.022 ± 0.010 mas) is resolved. Its mass is 0.783 ± 0.012, from asteroseismology with the radius measured by the interferometer. The same holds with 61 Cyg A (1.775 ± 0.013 mas, 3.49 pc, masses 0.67-0.74 from the orbits, which do not agree with each other). **With the registered instrumental classification, the sample does not support a mass limit of mass alone that grows with distance.** It is compatible with the reading by size. But it does not make it necessary.
- **Block Q, the state of the object.** No new target. The test is the prediction of section 5. The fraction f between the observed visibility and the one expected with the known decoherence is exactly the comparison with D1. One rule is added. D1 "is not enough" if an admitted paper gives $f + 2\sigma_f < 0.5$ (that is, more than half of the coherence lost, at 2σ). A mass above 10^{-21} kg and no known cause are also needed. A known cause is a cause given by the authors. It must have an estimate that accounts for at least half of the missing visibility. The quantity σ_f is the total uncertainty given by the authors. **Correction:** the rule had first been registered with the wrong sign, $f - 2\sigma_f$. The verifier found it. The rule was corrected before any paper could count.
- **Block W, a proposal: Wigner's friend with a heavy record.** Imagine a test of "local friendliness". In this test the record of each friend is a heavy mechanical mode, like the 16-microgram crystal of E3. It would decide between D4 (no mass limit if no copy goes out) and the readings at the Planck scale. These are readings T1 and T1b of the first part. The rule on the effective mass is used (a factor 100). Then the records should have at least 2.18 mg (T1) or 1.54 mg (T1b) of effective mass. To become a protocol it still lacks:
- a scheme with two laboratories;
- a way to make the mechanical mode record two outcomes and to cancel the recording;
- the measurements of the outside observers;
- a model of the leak of copies;
- the effect of the fidelities on the inequality.

The cat state of E3 (fidelity about 66 %) is a step, not yet a record of an outcome. No experiment with this combination was found. There are proposals and nearby experiments:

- Wigner's friend scenarios with macroscopic cat states;
- superpositions of heavy resonators;
- the program of "local friendliness" tests on quantum computers.

Deadline 31 December 2035, like section 5.

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6. What is not claimed

- No mass threshold is derived. And none has been found: no experiment has met one.
- It is not claimed that quantum mechanics is explained by the author's framework. Quantum mechanics predicts the probabilities by itself. In the framework the exponent 2 of the Born rule is taken from it (deposit 2.48).
- The compatibility of T2 and T4 with the data is coherence with certain facts. They are facts that quantum mechanics already predicts. It is not an independent test of the framework.
- The readings that do not follow from the data hit the translations of the two GCDs made by the AI, not the author's premise. The readings that remain (Planck scale, mass of the observer, Diósi-Penrose type) are not confirmed. They cannot be tested today.
- Take the reading at the Planck scale and the crystal of E3 (15.4 against 16.2 μg). Their closeness is not a result.
- The summary "MM in brief" writes of a closeness to Penrose and Diósi. After this work, that closeness is a possibility still to be tested. A favourable outcome would not distinguish the framework from Diósi-Penrose.
- Second part. The two rules of the fractional distance that hold are D1 for the state and D6g for the reading. They coincide with decoherence and with wave optics. The framework admits them, it does not predict them differently. D6g was written after the calculation. That the two rules are a single rule of the framework is a proposal.
- The fringes of starlight are a fact of the reading, not of the state of the star. It is not claimed that a star is in superposition. That Sirius B is not resolved is deduced from its radius and from the instrument. It is not a measured disc.
- The count of the photons of the stone on Mars says that the naked eye has no reliable reading. It does not say that the stone does not exist.
- No experiment changes only the distance between a system and whoever records it. And no Wigner's friend has a macroscopic mass. Those questions remain open.

How the work was done

Who did what. The author is not a physicist. The inventory, the translations, the programs and the calculations were made with an artificial intelligence (Claude, Anthropic) on 22 October 2026. A second artificial intelligence verified them (gpt-6-luna via Codex, OpenAI). It did one round for each level. The sentences of the author from which the work arises are in AUTHOR_BACKGROUND.md.

The remarks of the verifier and the errors found.

- **Level 1 (six remarks).** Accepted:
- the second value of R_0 found in two reviews ($4.94 \cdot 10^{-10}$ m) is not a limit on Diósi-Penrose. It is a CSL limit of the same paper, copied wrongly;
- the error bars of E5 are 95 % intervals, not standard deviations;
- E2 alone cannot exclude;
- the time of E3 is an asymptotic estimate.

Accepted in part: the sensitivity of T2 to the separation and to the choice between h and ħ, now declared (§4.3). No outcome changed.

- **Level 2 (six remarks).** Accepted:
- the program did not yet apply the admission rules (now it does);
- the margin of T3 was ambiguous. Now the thresholds apply to r with no margin, the factor 2 only in the rules on visibility;
- in the Planck block "favourable" could prevail over "excluded" (now it cannot);
- the rows of MAQRO at 100 s are study scenarios, not performances;
- T2 has the scale of free spreading only up to a factor 2π.
- **Errors found by the AI itself:**
- two internal checks written wrongly (corrected before any result);
- the meaning of T2, first presented as "coinciding with quantum mechanics" (corrected: it is a particular case, exceeded by atom interferometers);
- the parameters of MAQRO (corrected before any target could exist).
- The summary "MM in brief" uses the word "confirmation". Both rounds of the verifier found it stronger than what these data show. The summary is a text of the author. It was not changed here (§6).
- **The package** was verified by the same verifier, at most in two rounds. The first round (ten remarks) corrected these points:
- the summary of the rules of §5.2;
- the count of the readings that cannot be tested today (five, not two);
- a reference;
- a note on the translation of a sentence of the author;
- the list of the registers in the archive.

The second and last round found these corrections right. And it led to a more complete list of the contents in the Italian text.

- **The second part** (25 October 2026) was done with the same tools. The same verifier verified it in three runs: two rounds for Level 0, two for Level 1, one for Level 2. Every round said "not ready". Every correction was applied. The calculations were confirmed each time. The main ones:
- the experiments on distance are not a controlled series. So the sentence "no trend with distance" was withdrawn. And D0 became "not testable today";
- D4 and D6b moved to "not testable today". The reason is that the leak of copies and the number of copies are not measured;
- the sharp form of D6 does not follow from the data (HD 177724). And the graded form is declared as written after the calculation;
- the conclusion of block R was narrowed;
- the rule of block Q had the wrong sign;
- block W became a proposal.
- The extended package was then verified again. Round 3 on the package: five remarks, all accepted:
- the main result is now stated only for the two readings of distance. The reason is that the readings at the Planck scale of the first part remain untested;
- block R is stated with its restriction in every summary;
- D5a and D5b are kept distinct;
- the description says that block R was not a blind test. And it says that the non-resolution of Sirius B is deduced.

Round 4, the last, found no remarks and called the package ready. Reports and answers are in VERIFICA_PACCHETTO.

The registrations, with the times (UTC).

- The ten readings of the first part: 02_livello1/TRADUZIONI_LIVELLO1_v1.md, registered on 22 October 2026 at 12:39:50.
- The dated prediction: 05_livello2/PREVISIONE_LIVELLO2_v1.md, 22 October 2026 at 13:00:30. The valid rules are in CORREZIONE_2_PREVISIONE.md and in the program valuta_previsione_v2.py.
- The eight readings of the distance: 09_distanza_frazionale/02_livello1/TRADUZIONI_L1_v1.md, 25 October 2026 at 12:59:58. Programs: calcoli_L1.py (13:01:40) and calcoli_L1_v2.py (13:26:57, after the first verification).
- The tests of the distance: 09_distanza_frazionale/03_livello2/PREVISIONE_L2_v1.md, 13:08:06. The primary sources of block R were downloaded at 13:25:09. The program valuta_L2_R.py was registered at 13:27:55.

Where the data and the programs cited in the text are.

- Inventory: 01_livello0/INVENTARIO_LIVELLO0_v1.md, with the page numbers. For the distance: 09_distanza_frazionale/01_livello0.
- Fingerprints of the twenty papers: 00_fonti/IMPRONTE.txt (the papers are not redistributed).
- Calculations of the ten readings: 02_livello1/calcoli_livello1_v3.py. The results are in risultati_livello1.json and log_livello1.txt.
- Robustness of T2: 02_livello1/prova_robustezza.py. T2 beyond the Talbot geometry: 02_livello1/t2_oltre_talbot.py and 02_livello1/CORREZIONE_T2_LIVELLO1.md. Illustration: 02_livello1/illustrazioni.py.
- Valid table of the projects: 05_livello2/log_progetti_v2.txt. The tables in log_previsione.txt and log_previsione_v2.txt use the earlier, wrong parameters of MAQRO. The 45 tests of the program: log_previsione_v2.txt. The program is used by importing valuta_previsione_v2.py. Then you call valuta_P() or valuta_G() with the numbers of the paper.

How to reproduce. There are seven programs in Python 3, with no external libraries. They run in a few seconds in all:

- in 02_livello1:
- calcoli_livello1_v3.py (the registered calculation of the ten readings);
- illustrazioni.py, prova_robustezza.py and t2_oltre_talbot.py (the illustration and the checks after the calculation);
- in 05_livello2:
- valuta_previsione.py (the prediction);
- progetti_v2.py (the table of the projects);
- valuta_previsione_v2.py (the rules after correction 2).

Every protocol and every program has a SHA-256 fingerprint and a UTC time registered before the calculation. The results are in JSON files and in the logs. Two earlier versions of the first program (calcoli_livello1.py and _v2.py) stopped at an internal check that was wrong. This happened before producing any result (DEVIAZIONI_LIVELLO1.md and _D3.md). They are kept for the record. The search on arXiv (controllo_arxiv.py) and the extraction of the text from the papers (estrai_testo.py) are not re-run. The first depends on the day on which it is run. The second needs the papers, which are not redistributed.

For the second part there are three more programs in 09_distanza_frazionale, with no external libraries:

- calcoli_L1.py (the registered calculation of the eight readings);
- calcoli_L1_v2.py (the corrections after the first verification);
- 03_livello2/valuta_L2_R.py (block R).

Ten programs in all. The two tools that extract and search the text of the sources are 00_fonti_estrai_testo.py and 00_fonti_cerca.py. They are not re-run. They need the papers.

The command is:

python run_all.py [output_folder]

The script re-runs the ten programs in a new or empty folder. If none is given, it uses a temporary folder. Then it

compares the results with the registered ones. It compares the JSON files number by number, the logs line by line. The text must be identical. The numbers are compared with a relative tolerance of 10^{-9} of the larger value, with no absolute minimum. The line-ending characters and the empty lines at the end of a log are not compared. The script ends with "ALL MATCH". It is tested with Python 3.13.13 (the version is written in the record of the test from scratch). No other libraries are needed. It takes a few seconds.

Contents of the package.

- README.md (the same text in English); LEGGIMI_IT.md (the Italian text); AUTHOR_BACKGROUND.md (the sentences of the author, kept out of the main text); LICENSE.md (texts CC BY 4.0, code MIT).
- 07_soglia_massa/01_livello0:
- the inventory of the data (version 1, values read in the papers, with the page numbers);
- its corrections after the verification;
- the program that extracted the text from the papers.
- 07_soglia_massa/00_fonti: IMPRONTE.txt, the fingerprints of the 20 papers downloaded from arXiv (the papers are not redistributed; they are in the bibliography).
- 07_soglia_massa/02_livello1:
- the ten readings with the outcome rules (registered before the calculation);
- the three versions of the program and the two notes on the deviations;
- the results (JSON and log);
- the checks after the verification;
- the program on T2 beyond the Talbot geometry;
- the correction on T2;
- the illustration;
- the three outcome notes (v3 is the final one).
- 07_soglia_massa/05_livello2:
- the search on arXiv before the prediction;
- the dated prediction and its program;
- the two corrections with their programs and results;
- the final note.
- 07_soglia_massa/03_verifica_astra: the two verification reports on the calculations (gpt-6-luna via Codex) and the answers; VERIFICA_PACCHETTO: the reports of the same verifier on the package and the answers.
- 07_soglia_massa/04_riquadro: only REGISTRO.txt. It contains the fingerprints and the times of the proposals made to the author for the sentence of the summary "MM in brief". The proposals are not part of this deposit.
- 09_distanza_frazionale, the second part. It has a single REGISTRO.txt for the whole folder, with the relative paths. It contains:
- the two first evaluations of the author's remarks (NOTA_VALUTAZIONE_v1 and v2, written before the protocol: contamination);
- 00_fonti, the fingerprints of the 22 sources downloaded with the author's permission (not redistributed) and the two tools for the text;
- 01_livello0, the inventory of the book and of the deposits, the inventory of the experiments, its correction and the outcome note;
- 02_livello1, the eight readings with the outcome rules (registered before the calculation), the two programs with results and logs, the three outcome notes (v3 is the final one);
- 03_livello2, the tests, the program of block R with the results, the two outcome notes (v2 is the final one);
- 04_verifica_astra, the three reports of the verifier.
- Left out of the archive, for the second part:
- the two remarks of the author with the exact words (they are in AUTHOR_BACKGROUND.md);
- the messages to the verifier and the programs that composed them;
- the report written for the author and its correction;
- the sources and their extracted text.

The fingerprints are in the REGISTRO.txt of 09_distanza_frazionale.

- REGISTRO.txt in every folder (and REGISTRO_PACCHETTO.txt, REGISTRO_TESTI.txt for the texts and the programs of the package). They contain the SHA-256 fingerprint and the UTC time of every registration. Some registered working files are not in the archive:
- the author's premise with the exact words;
- the provisional version of the inventory (with the list of the downloads asked of the author);
- the messages to the verifier (which contain all the material) and the programs that composed them;
- the proposals for the sentence of the summary "MM in brief".

Their fingerprints are in the REGISTRO.txt files. They can be shown on request.

- run_all.py re-runs the calculations and compares them with the registered ones. IMPRONTE.sha256 contains the fingerprints of all the other files.
- In the working records this deposit is called "soglia di massa" (folder MM_sviluppo_quantistica/07_soglia_massa). The second part is called "distanza frazionale" (folder 09_distanza_frazionale). The names of the files and the registered documents remain in Italian.

Related records. 2.64 (DOI 10.5281/zenodo.23610071), which announced this work. 2.40 (DOI 10.5281/zenodo.23280168), the clock as a geometric mean (reading T2). 2.38 (DOI 10.5281/zenodo.23055978), the layers (readings T1, T1c). 2.48 (DOI 10.5281/zenodo.23352814), Born rule and Tsirelson bound. The status reports 2.0-2.27 (concept DOI 10.5281/zenodo.21802475), with the two-walls invariant (2.9-2.18) and the August note on the threshold (2.9). For the second part:

- 2.62 (DOI 10.5281/zenodo.23609967), objects as the common part of two points of view, the ruler in fractions and the threshold of copies (readings D1, D4, D6c);
- 2.36.2 (DOI 10.5281/zenodo.23287965), distance as the fragmentation of a register (readings D5a, D5b);
- 2.50 (DOI 10.5281/zenodo.23376729), an earlier translation of «venuta in essere e distanza di diffrazione» ("coming into being and diffraction distance").

This version. This student version was written with an AI assistant (Claude) from the archived Italian text (LEGGIMI_IT.md). The numbers were checked with a program against the archived text. This English text is a translation of the Italian student version, which the author read and approved.

Glossary

- **a_0:** the acceleration scale of galaxies, about $1.2 \cdot 10^{-10}$ m/s². Below this acceleration the rotation of galaxies departs from Newton's law.
- **Atomic mass unit (u):** $1.66 \cdot 10^{-27}$ kg, about the mass of a proton; 1 kDa = 1,000 u, 1 MDa = 10^6 u.
- **Cat state:** a superposition of two clearly different states of a large system.
- **Centre of mass:** the point that moves as if all the mass were there. A superposition of the centre of mass means that the whole object is in two places.
- **Collapse model:** a modification of quantum mechanics that makes superpositions end by themselves. It makes them end faster for larger objects (GRW, CSL, Diósi-Penrose).
- **Compton wavelength:** h/(mc), the length linked to a mass m. The reduced one is ħ/(mc).
- **Contamination:** what was known of the result when a rule was written.
- **Copy:** in the author's framework, each photon or molecule that carries away a trace of where an object is.
- **CSL:** continuous spontaneous localisation, the continuous version of the GRW model. It has a rate λ and a length r_C.
- **Decisive, supporting, not decisive:** the three groups into which the experiments were put before the calculation.
- **Decoherence:** the loss of the stripes caused by the environment, which records where the object is.
- **Diffraction cell:** the smallest angle that an aperture of size a can tell apart, about 1.22 λ/a. Multiplied by the distance, it gives the smallest size that can be told apart there.
- **Diósi-Penrose model:** a superposition of two positions of a mass lasts about ħ/E_G. There is a version with a radius R_0 for the nuclei.
- **Effective mass:** the mass assigned to a vibration mode. It depends on how the authors define its amplitude.
- **E_G:** the gravitational energy of the difference between the two arrangements of the mass in a superposition.
- **Fractional distance:** a term of the author. In this work, the size to be told apart measured in the cell of the copies of whoever reads (§3.3).
- **GCD:** greatest common divisor. In the author's framework it is the only logical element that space and time have in common. "Principal GCD" and "GCD of the diffraction" are terms of the framework (§2.2).
- **GRW:** the collapse model of Ghirardi, Rimini and Weber (1986).
- **Interferometer:** an apparatus that splits a wave and recombines it, so that the parts can interfere. A Talbot-Lau interferometer uses three gratings.
- **Local friendliness:** inequalities that hold if the outcomes observed by "friends" in closed laboratories are absolute, the agents are free and the influences are local. Quantum mechanics violates them (Bong and others 2020).
- **Macroscopicity μ:** a number that says how much an experiment restricts the modifications of quantum mechanics of the collapse-model type (Nimmrichter and Hornberger 2013).
- **MAQRO:** a proposed space mission for interference with particles from 10^7 to 10^{11} u.
- **mas:** milliarcsecond, $4.85 \cdot 10^{-9}$ rad.
- **Planck mass (m_P):** $\sqrt{(\hbar c/G)} = 2.18 \cdot 10^{-8}$ kg, about 22 micrograms.
- **Ratio r:** for each reading and experiment, how far the experiment is from the threshold of the reading; $r < 1$ on the quantum side.
- **Reading (L) and state (Q):** how an object appears in the record of an observer, and what the object itself does.

The second part keeps them apart.

- **Registered:** written in a file whose SHA-256 fingerprint and UTC time were noted before the calculation. The file is not changed afterwards.
- **Resolved:** an object whose angular size an instrument can measure. Unresolved objects appear as the diffraction figure of the instrument.
- **Schwarzschild radius:** $2Gm/c^2$, the radius at which a mass m would become a black hole.
- **Standard deviation (σ):** the measure of statistical uncertainty; 3σ corresponds to a confidence of 99.73 %.
- **Superposition:** the state of an object spread over several positions (or states) at the same time.
- **Talbot condition:** the condition under which a three-grating interferometer forms the stripes. Between two gratings the wave must spread at least as much as the period of the grating.
- **van Cittert-Zernike theorem:** the light of a distant, incoherent source of angular diameter θ is coherent over a transverse distance of about λ/θ. For a uniform disc the visibility of the fringes with baseline B is $|2J_1(x)/x|$, $x = \pi\theta B/\lambda$.
- **Visibility:** the contrast of the stripes, from 0 (none) to 1.
- **Wigner function:** a description of a quantum state in position and momentum. Where it is negative, no classical description is possible.
- **Wigner's friend:** a thought experiment, today carried out with photons and qubits. An observer (the friend) measures inside a closed laboratory. An outside observer treats the laboratory as a quantum system.

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